

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of $arr[i] * i$, where i is the index of the element ($i = 0, 1, 2, \dots, N$). Write an algorithm based on Greedy technique with a Complexity $O(n \log n)$.

Input Format:

First line specifies the number of elements- n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int compare(const void *a, const void *b) {
5     return (*(int *)a - *(int *)b);
6 }
7
8 int main() {
9     int n;
10    scanf("%d", &n);
11
12    long long arr[n];
13
```

```

13
14 ▼ for (int i = 0; i < n; i++) {
15     scanf("%lld", &arr[i]);
16 }
17
18 qsort(arr, n, sizeof(long long), compare);
19
20 long long sum = 0;
21
22 ▼ for (int i = 0; i < n; i++) {
23     sum += arr[i] * i;
24 }
25
26 printf("%lld", sum);
27
28 return 0;
29 }

```

	Input	Expected	Got	
✓	5	40	40	✓
	2			
	5			
	3			
	4			
	0			